Unsupervised Learning Laboratories

Lab 3

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3 Isomap

In Lab 2 we saw that PCA, being a linear projection, cannot unroll the Swiss Roll. Isomap replaces the Euclidean distances with *geodesic* distances estimated on a neighbourhood graph, and then applies classical MDS.

You can use external libraries for linear algebra operations (e.g. `numpy.linalg`) but you are expected to write your own algorithms, including the computation of the shortest paths.

3.1 Exercise 1

- Write your own implementation of Isomap.
- Apply it to the Swiss Roll dataset ($n = 1000$) from Lab 1, and colour the points by φ. Try different values of the number of neighbours k and comment on how this choice affects the result.
- Use a modified version of the Swiss Roll dataset, in which Gaussian noise from a Normal $\mathcal{N}(\mu = 0, \sigma = 0.5)$ is added to the x and y coordinates. Apply Isomap to this dataset and discuss the differences with the previous point.

3.2 Exercise 2

Undersample randomly from the Dry Bean Dataset in order to have $n = 1000$ datapoints. Follow the same pipeline of Exercise 1 of the previous lab by replacing PCA with Isomap. Discuss the differences with particular focus on the accuracy of the logistic regression.

Important note: in this case you don't need to do the train-test split first. Apply Isomap to the whole dataset, then do the train-test split and follow the rest of the pipeline.

Notes

- You can use the `sklearn.neighbors.NearestNeighbors` class.
- You are expected to write your own implementation of the Floyd-Warshall algorithm, as seen in class. If you wish, you can implement Dijkstra's algorithm instead.
- Remember that your distance matrix must be symmetric!
- Suggestion: test your algorithm with $n = 100$ points at first.

Bonus

Study how the accuracy of the logistic regression on the Dry Bean dataset depends on the number of neighbours k.